

SIMATS ENGINEERING
CO4 - ASSESSMENT TOOL 3
Article Review

COURSE NAME: Deep Learning **COURSE CODE:** MLA0408

COURSE OUTCOME COVERED

CO 4: Students will be able to analyze and evaluate advanced deep learning techniques and architectures, interpret research findings, and apply appropriate models to real-world problems. They will be able to critically review research articles, compare methodologies, and justify suitable deep learning solutions. (BL-6)

Assessment Tool Weightage: 30%

QUESTIONS:

Q1. Article Identification and Summary

Select a recent peer-reviewed research article related to Deep Learning, such as CNNs, Transfer Learning, ResNet, DenseNet, RNNs, LSTM, Autoencoders, or other advanced architectures. Provide the article title, authors, publication venue, year, and research objective. Summarize the main problem addressed, proposed approach, and key findings in your own words.

Q2. Problem Statement and Motivation

Explain the real-world problem addressed by the selected article and analyze why the problem requires a deep learning solution. Identify the limitations of existing approaches discussed by the authors and explain the motivation for the proposed method.

Q3. Methodology and Architecture Review

Critically review the methodology used in the article. Describe the dataset, preprocessing, model architecture, important layers or components, training strategy, and evaluation metrics. Illustrate the proposed architecture or workflow and explain the role of its major components.

Q4. Results, Comparison and Critical Analysis

Analyze the experimental results reported in the article. Compare the proposed method with the baseline or existing methods using the reported performance measures. Discuss the strengths, limitations, computational considerations, and validity of the results. Identify at least two possible improvements or future research directions.

Q5. Application and Personal Evaluation

Explain how the reviewed deep learning method can be applied to a practical industry problem. Propose a suitable implementation or extension and justify your choices. Conclude the review with your overall evaluation of the article, including its contribution, practical relevance, and scope for future work.

Article Review Submission Guidelines

- Choose one recent and relevant peer-reviewed research article in Deep Learning.
- Attach or provide the complete citation/reference of the selected article.

- The review must be written in the student's own words and should demonstrate critical analysis.
- Include architecture diagrams, tables, graphs, or screenshots where they improve explanation.
- Use appropriate academic referencing and avoid plagiarism.
- Recommended review length: 4–6 pages, excluding the reference page.

Code of Conduct:

I D.BINDUPRIYA(192425162) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Bindu Priya

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Understanding & Summary	20%	Comprehensive and accurate understanding of the article, problem, objectives, and contributions.	Good understanding with minor omissions or inaccuracies.	Basic understanding; important details are missing.	Poor understanding or significant misinterpretation.
Critical Review of Methodology	25%	Thoroughly evaluates methodology, architecture, datasets, training, and evaluation choices.	Appropriate analysis with minor gaps.	Limited analysis with several unexplained aspects.	Little or no meaningful analysis of methodology.
Analysis of Results	20%	Accurately interprets results, comparisons, metrics, strengths, and limitations.	Good interpretation with reasonable comparison.	Basic interpretation with limited comparison.	Incorrect, unsupported, or missing interpretation.
Application & Proposed Improvements	20%	Practical, innovative application with well-justified improvements and future directions.	Relevant application and feasible improvements.	Basic application with limited justification.	Unclear, impractical, or unsupported proposal.
Presentation & Referencing	15%	Well-organized, professional presentation with	Clear presentation with minor formatting or	Adequate presentation with inconsistent	Poor organization, unclear

		clear diagrams and correct academic referencing.	referencing issues.	formatting/referen cing.	presentation, or inadequate referencing.
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Suggested Article Review Structure

1. Article Citation
2. Abstract / Executive Summary
3. Research Problem and Motivation
4. Methodology and Deep Learning Architecture
5. Dataset and Experimental Setup
6. Results and Comparative Analysis
7. Strengths and Limitations
8. Proposed Improvements / Future Scope
9. Practical Industry Application
10. Conclusion and References

Selected Research Article

Deep Neural Networks with Transfer Learning for Medical Image Classification (ResViT-MedNet)

Authors: Gajanan Joshi

Journal: *EPJ Web of Conferences (ICATCICT 2025)*

Year: 2025

Volume: 341

Article: 01040

Publication Date: 2025

Publisher: EDP Sciences

DOI: 10.1051/epjconf/202534101040

1. ARTICLE CITATION

Citation

Joshi, G. (2025). Deep Neural Networks with Transfer Learning for Medical Image Classification. EPJ Web of Conferences, 341, Article 01040.

Research Area

Deep Learning → Convolutional Neural Networks → Vision Transformers → Transfer Learning → Medical Image Classification

Main Topic

The research focuses on a hybrid CNN–Transformer transfer learning model, named ResViT-MedNet, that combines a ResNet-50 backbone pretrained on ImageNet with a Vision Transformer encoder and an attention-based fusion module to classify dermoscopic skin lesion images as malignant or benign.

Main Objectives

The research has two major objectives:

1. Design a hybrid architecture that combines the local, texture-level feature extraction strength of CNNs with the global, contextual dependency modelling of Vision Transformers.
2. Demonstrate, on a large real-world dermoscopic image dataset, that this hybrid model classifies skin lesions more accurately and with better interpretability than standalone CNN or Transformer baselines.

The study therefore goes beyond conventional single-architecture transfer learning by fusing two complementary feature-extraction paradigms into one interpretable classification pipeline.

2. ABSTRACT / EXECUTIVE SUMMARY

Medical image classification plays a crucial role in clinical diagnosis and disease screening, but deep learning models applied to this domain typically face two recurring problems: a shortage of large, expertly labelled datasets, and limited interpretability, since most convolutional models behave as black boxes that focus only on local spatial patterns rather than global contextual relationships. The selected

research proposes ResViT-MedNet, a Hybrid CNN–Transformer Transfer Learning Model that addresses both problems at once.

The architecture uses a ResNet-50 backbone, pretrained on ImageNet, to extract local, hierarchical feature maps capturing fine-grained patterns such as pigment networks and lesion margins. In a parallel branch, the same image is divided into patches, embedded, and passed through a stack of Transformer encoder layers that use multi-head self-attention to model long-range, global relationships between distant regions of the lesion. An attention-based fusion module then combines the local CNN features and the global Transformer embeddings into a single, richer feature representation, which is further refined using a channel-wise attention mechanism before being passed to a fully connected classification head.

The model was evaluated on the ISIC 2020 Melanoma Classification Dataset, containing 33,126 dermoscopic images labelled as malignant or benign. ResViT-MedNet achieved an accuracy of 98.47%, precision of 0.982, recall of 0.984, F1-score of 0.983, and an AUC of 0.994, outperforming baseline architectures including VGG16, ResNet50, ViT-Base, and EffNet-ViT. The results indicate that combining local and global feature extraction through attention-based fusion produces a more discriminative and better-generalizing model than either a pure CNN or a pure Transformer approach used alone.

3. RESEARCH PROBLEM AND MOTIVATION

3.1 Research Problem

Skin cancer, particularly melanoma, is one of the most commonly diagnosed cancers worldwide, and early, accurate classification of dermoscopic images is critical for timely treatment. Several challenges make automated skin lesion classification difficult:

- Malignant and benign lesions can appear visually similar, especially in early stages.
- Expert-annotated dermoscopic datasets are expensive and time-consuming to produce, leading to data scarcity.
- Standard CNN-based models are effective at capturing local texture but have a receptive field that grows slowly with depth, limiting their ability to model long-range dependencies across a lesion.
- Many existing transfer-learning-based CNN models offer limited interpretability, which reduces clinician trust in automated predictions.
- Attention-based interpretability methods that could address this gap remain relatively under-explored in medical imaging pipelines.

There is therefore a need for a model that can jointly capture fine-grained local texture and global contextual structure in a dermoscopic image, while remaining data-efficient and reasonably interpretable.

3.2 Traditional Disease Detection

Traditional skin lesion assessment mainly depends on: Dermoscopic Examination → Visual Inspection by Dermatologist → Manual Pattern Analysis (ABCDE rule, texture, colour) → Diagnosis. This

approach is effective in the hands of an experienced dermatologist but is subjective, time-consuming, and not easily scalable to population-level screening programs.

3.3 Why Deep Learning?

Deep learning is well suited to dermoscopic image analysis because lesion images contain complex, high-dimensional visual patterns that are difficult to fully capture with hand-crafted descriptors such as local binary patterns or histograms of oriented gradients.

The model can learn features such as: lesion border irregularity, colour variation, pigment network structure, texture asymmetry, and the spatial relationship between different regions of the lesion. Instead of manually engineering these descriptors, a deep learning model can learn useful, task-specific representations directly from labelled training images.

3.4 Motivation for a Hybrid CNN–Transformer Design

Convolutional layers are highly effective at extracting local, spatially correlated patterns because convolutional kernels act as sliding feature detectors, but their receptive field grows only gradually with network depth, making it harder for a purely convolutional model to relate features across distant parts of an image. Vision Transformers address this limitation directly, since self-attention computes a relationship between every pair of patches in the image regardless of their distance.

For example:



However, Transformers typically require large labelled datasets to train effectively and can underperform on fine-grained, low-level detail compared to CNNs. The authors therefore motivate a hybrid design that keeps a CNN backbone for efficient local feature extraction while adding a Transformer branch purely for global context modelling, fusing the two through a learnable, attention-based mechanism rather than relying on either architecture alone.

4. METHODOLOGY AND DEEP LEARNING ARCHITECTURE

4.1 Overall Workflow

The proposed system follows a two-branch pipeline. A dermoscopic image is first preprocessed and augmented, and then passed simultaneously through a CNN branch and a Transformer branch. The CNN branch, built on a ResNet-50 backbone pretrained on ImageNet, extracts hierarchical local feature maps. The Transformer branch divides the same image into patches, embeds them with positional encoding, and processes the sequence through stacked Transformer encoder layers to obtain globally contextualized embeddings. An attention-based fusion module combines the two feature streams, and the fused representation is refined with channel-wise attention before being passed to a fully connected classification head that outputs the final malignant-versus-benign prediction.

Figure 1: Overall Workflow of the Proposed ResViT-MedNet System for Skin Lesion Classification

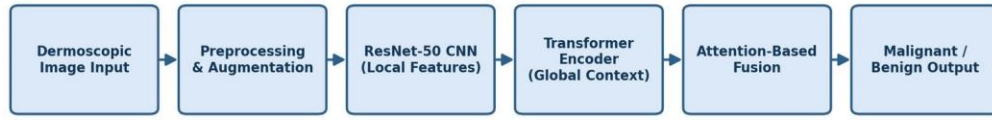


Figure 1: Overall Workflow of the Proposed ResViT-MedNet System for Skin Lesion Classification

4.2 Preprocessing and Augmentation

Every dermoscopic image is resized to 224 x 224 pixels so that it is compatible with the ImageNet-pretrained CNN backbone. Pixel values are normalized using the standard ImageNet channel statistics, subtracting the per-channel mean and dividing by the per-channel standard deviation, so that the input distribution matches what the pretrained backbone was originally trained on. To improve generalization and reduce overfitting, the training images are augmented using random horizontal and vertical flips, rotation up to plus or minus thirty degrees, scaling between 0.8x and 1.2x, brightness adjustment of up to plus or minus twenty percent, and small translations of up to ten pixels. These augmentations make the model more robust to variations in lesion orientation, illumination, and framing that naturally occur across different dermoscopic imaging devices and clinical settings.

4.3 CNN Backbone for Local Feature Extraction

The CNN branch of ResViT-MedNet uses a ResNet-50 backbone, pretrained on ImageNet, to convert the input dermoscopic image into a set of hierarchical feature maps. Convolutional kernels slide across the image, acting as local feature detectors, and successive residual blocks progressively capture increasingly abstract patterns, from simple edges in early layers to complex textures such as pigment networks and lesion margins in deeper layers. Because the backbone is pretrained on a large natural-image dataset and then adapted to the target dermoscopic dataset through transfer learning, the model can achieve strong performance even though the labelled skin lesion dataset is comparatively small.

4.4 Patch Embedding and Transformer Encoder

In parallel with the CNN branch, the Transformer branch splits the image-derived feature map into a sequence of flattened patches. Each patch is projected into an embedding vector using a learnable linear transformation, and a positional encoding is added so the model retains information about each patch's original location. This sequence of patch embeddings is processed by a stack of Transformer encoder layers, each consisting of Multi-Head Self-Attention followed by a feed-forward network, with residual connections and layer normalization used throughout. The self-attention mechanism computes attention scores between every pair of patches, allowing the model to learn relationships between spatially distant regions of the lesion, such as correlations between opposite edges of an irregular border. The output of the final encoder layer forms a set of globally contextualized embeddings.

4.5 Attention-Based Feature Fusion

The local features from the CNN branch and the global contextual embeddings from the Transformer branch are combined using an attention-based fusion module. The Transformer embeddings are passed through a small multi-layer perceptron and then combined with the CNN feature maps using a learnable weighting parameter that balances the relative contribution of local and global information. A channel-

wise attention block further recalibrates the fused features, using global average pooling followed by a small bottleneck of fully connected layers, so that the most diagnostically relevant feature channels are emphasized before classification.

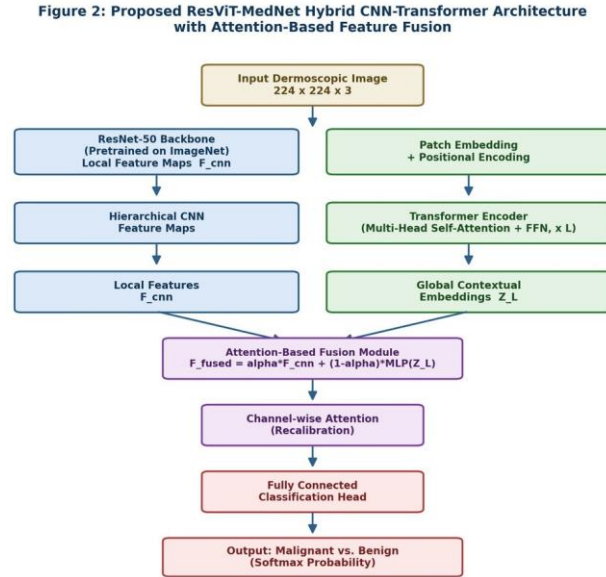


Figure 2: Proposed ResViT-MedNet Hybrid CNN-Transformer Architecture with Attention-Based Feature Fusion

4.6 Classification Head

The fused and channel-recalibrated feature representation is flattened and passed through fully connected layers that produce the final class probabilities using a softmax activation function. The model is trained end to end by minimizing the categorical cross-entropy loss between the predicted probabilities and the ground-truth malignant or benign labels, allowing the CNN backbone, Transformer encoder, and fusion module to be jointly optimized for the classification task.

5. DATASET AND EXPERIMENTAL SETUP

5.1 Dataset

The researchers used the ISIC 2020 Melanoma Classification Dataset, a large, publicly available benchmark for skin lesion analysis.

The dataset contains:

- 33,126 dermoscopic images labelled as malignant or benign
- Images compiled from multiple international clinical sources
- Metadata accompanying each image, including lesion type, patient sex, age, and patient ID

This large, diverse, and clinically sourced dataset makes ISIC 2020 a widely used and challenging benchmark for evaluating deep learning models on real-world dermoscopic image classification.

5.2 Dataset Split

Split	Proportion	Method
Training	70%	Patient-

		level stratified split
Validation	15%	Patient-level stratified split
Testing	15%	Patient-level stratified split

A patient-level stratified split was used specifically to avoid data leakage between the training, validation, and testing sets, since multiple images can originate from the same patient.

5.3 Training Configuration

Parameter	Value
CNN backbone	ResNet-50 (pretrained on ImageNet)
Global feature extractor	Transformer Encoder (Multi-Head Self-Attention + FFN)
Fusion mechanism	Learnable attention-based fusion + channel-wise attention
Input image size	224 x 224 x 3
Loss function	Categorical Cross-Entropy
Augmentation	Flips, rotation ($\pm 30^\circ$), scaling (0.8–1.2x), brightness ($\pm 20\%$), translation ($\pm 10\text{px}$)

5.4 Evaluation Metrics

The performance of ResViT-MedNet and the baseline models is evaluated using several standard classification metrics. Accuracy measures the overall proportion of correctly classified dermoscopic images. Precision measures how many of the images predicted as malignant were actually malignant, while recall measures how many of the truly malignant images were correctly identified, which is particularly important in a screening context where missed malignant cases carry serious consequences. F1-score provides a balanced measure between precision and recall, and the Receiver Operating Characteristic (ROC) curve, together with its Area Under the Curve (AUC), is used to visualize how well the model separates malignant from benign cases across different decision thresholds. A confusion matrix is also used to examine class-wise performance and to compare how often each model confuses benign and malignant lesions.

6. RESULTS AND COMPARATIVE ANALYSIS

6.1 Overall Performance of ResViT-MedNet

The proposed ResViT-MedNet model achieves an overall accuracy of 98.47%, with a precision of 0.982, a recall of 0.984, an F1-score of 0.983, and an AUC of 0.994 on the ISIC 2020 test set. The confusion matrix reported in the article shows strong diagonal dominance, meaning that the large majority of malignant and benign predictions fall on the correct classification line, with comparatively few false positives and false negatives. The close agreement between precision and recall indicates that the model maintains a good balance between sensitivity and specificity rather than favoring one at the expense of the other.

6.2 Comparative Analysis

The authors compare ResViT-MedNet against four baseline architectures: VGG16, ResNet50, ViT-Base, and EffNet-ViT. The reported results can be summarized as follows:

Model	Accuracy	Precision	Recall	F1-Score	AUC
VGG16	93.24%	0.921	0.918	0.919	0.954
ResNet50	94.12%	0.938	0.935	0.936	0.963
ViT-Base	95.37%	0.950	0.946	0.948	0.972
EffNet-ViT	96.21%	0.961	0.959	0.960	0.982
ResViT-MedNet (Proposed)	98.47%	0.982	0.984	0.983	0.994

ResViT-MedNet outperforms every baseline on all five metrics, achieving over 2.2% higher accuracy than EffNet-ViT and close to 5% higher accuracy than VGG16, along with the highest AUC of 0.994 among all compared models. The authors attribute this improvement to the combination of local CNN feature extraction with global Transformer context modelling through the attention-based fusion module, which allows the model to capture both fine-grained texture and long-range structural relationships that neither a pure CNN nor a pure Transformer captures as effectively on its own.

7. STRENGTHS AND LIMITATIONS

7.1 Strengths

The proposed ResViT-MedNet architecture has several notable strengths. Combining a pretrained ResNet-50 backbone with a Transformer encoder allows the model to benefit from strong local feature extraction while also modelling long-range dependencies that convolutional layers alone struggle to capture. The use of transfer learning from ImageNet significantly reduces the amount of labelled dermoscopic data required for effective training, which is valuable given the cost and scarcity of expert-annotated medical images. The attention-based fusion mechanism, together with the channel-wise attention block, provides a degree of interpretability by allowing the relative contribution of local and global features to be inspected, which can help clinicians understand which aspects of an image most influenced a prediction. The comprehensive comparison against four strong baseline architectures,

evaluated using accuracy, precision, recall, F1-score, and AUC, gives confidence that the reported performance gains are genuine rather than an artefact of a narrow evaluation.

7.2 Limitations

1. **Single Dataset Evaluation:** The model is validated only on the ISIC 2020 dataset, and its performance on other dermoscopic datasets or on entirely different imaging modalities remains untested.
2. **Binary Classification Scope:** The current study addresses only malignant-versus-benign classification, rather than distinguishing between multiple specific skin disease categories, which would be more clinically informative.
3. **Computational Cost:** Combining a ResNet-50 backbone with a Transformer encoder increases the number of parameters and computational requirements compared to a single-branch CNN or Transformer model, which the article does not discuss in detail.
4. **Limited Quantitative Interpretability Analysis:** While attention-based fusion offers some interpretability, the article does not include a formal, quantitative evaluation of how well the attention weights align with regions a dermatologist would consider diagnostically important.
5. **Generalization Across Skin Tones and Devices:** The article does not report a detailed breakdown of performance across different skin tones, imaging devices, or clinical sites, which is an important consideration for real-world deployment.

8. PROPOSED IMPROVEMENTS / FUTURE SCOPE

8.1 Multi-Class Disease Classification

Extending ResViT-MedNet from binary malignant-versus-benign classification to multi-class classification across several specific skin disease categories, such as melanoma, basal cell carcinoma, and benign nevi, would make the system considerably more useful for real diagnostic decision-making, since clinicians typically need to know the specific condition rather than only a binary risk flag.

8.2 Multimodal Imaging Data

Future work could extend the model to incorporate multimodal data, such as combining dermoscopic images with patient metadata (age, sex, lesion location) or with additional imaging modalities. This could allow the fusion module to combine visual features with clinically relevant contextual information, potentially further improving diagnostic accuracy.

8.3 Self-Supervised Pretraining

Rather than relying solely on ImageNet pretraining, future versions of the model could use self-supervised pretraining directly on large collections of unlabelled dermoscopic images. This could help the CNN and Transformer branches learn representations that are more directly relevant to skin lesion characteristics, potentially improving performance further while reducing dependence on labelled natural-image datasets.

8.4 Explainability Validation

A lightweight, quantitative study could formally evaluate the attention-based fusion weights and channel-wise attention maps against regions that dermatologists identify as diagnostically important, for example using the ABCDE criteria (Asymmetry, Border, Colour, Diameter, Evolution). This would

move the model's interpretability claims from a qualitative discussion toward a measurable, clinically validated property.

9. PRACTICAL INDUSTRY APPLICATION

The reviewed method can be applied to clinical decision support and large-scale dermatological screening.

9.1 Clinical Decision Support Tool

A dermatology clinic or hospital could integrate ResViT-MedNet into its imaging workflow so that every captured dermoscopic image is automatically screened before a dermatologist reviews it.

Dermoscopic Image Capture → Preprocessing → CNN + Transformer Feature Extraction → Attention-Based Fusion → Malignant/Benign Prediction → Dermatologist Review

The system could flag high-risk lesions for urgent review and display a confidence score alongside the prediction, helping prioritize which patients are seen first in a busy clinical setting.

9.2 Teledermatology and Remote Screening

In regions where dermatologists are not readily available, patients or general practitioners could capture a dermoscopic image using a portable dermatoscope attached to a smartphone, and the model could provide an initial automated risk assessment, helping determine whether an urgent referral to a specialist is warranted.

9.3 Population-Level Screening Platform

The proposed architecture could be extended into a broader skin-screening platform used in community health camps or workplace wellness programs, where large numbers of dermoscopic images are captured and automatically triaged using ResViT-MedNet. Images flagged as high risk could be forwarded for specialist review, while low-risk images could be logged for routine monitoring. Over time, such a platform could combine image-based screening with patient history to support earlier detection of skin cancer at a population scale, reducing the burden on dermatology departments while improving overall screening coverage.

10. CONCLUSION AND REFERENCES

10.1 Overall Evaluation

The selected article presents a well-designed hybrid CNN–Transformer architecture that directly addresses two persistent problems in medical image classification: limited labelled data and limited interpretability. Its major contribution lies in the attention-based fusion module, which combines the local feature extraction strength of a pretrained ResNet-50 backbone with the global contextual modelling of a Transformer encoder, rather than treating the two architectures as competing alternatives. The reported accuracy of 98.47% and AUC of 0.994 on the ISIC 2020 dataset are strong results, and the comparison against four increasingly sophisticated baselines, VGG16, ResNet50, ViT-Base, and EffNet-ViT, provides convincing evidence that the hybrid design offers a genuine advantage rather than a marginal improvement.

In my evaluation, the research is technically sound and practically relevant, since skin cancer screening is a high-impact clinical problem and the proposed fusion strategy is a reasonable and well-motivated

way to combine complementary feature-extraction paradigms. However, before such a model could be relied upon in real clinical deployment, it would benefit from validation on additional, independent dermoscopic datasets, extension to multi-class disease classification, and a more rigorous, quantitative evaluation of its interpretability claims.

10.2 Personal Evaluation

From my perspective, the most valuable aspect of this research is that it does not simply pick one architecture and hope it works; it explicitly recognizes the complementary strengths of CNNs and Transformers and builds a mechanism to combine them in a learnable way. This is a more thoughtful design choice than reflexively defaulting to whichever architecture is most fashionable, and it produces a model that is both accurate and reasonably interpretable.

I would improve the system by extending it to multi-class skin disease classification rather than binary malignant-versus-benign prediction, since this would provide far more clinically actionable information. I would also evaluate the model on additional dermoscopic datasets collected from different devices and patient populations, and conduct a formal study comparing the model's attention maps against dermatologist-annotated regions of interest. Overall, the research provides a strong and practically motivated foundation for building accurate, interpretable AI systems for dermatological screening.

10.3 Final Conclusion

The research demonstrates the effectiveness of combining transfer-learning-based CNNs with Vision Transformers, through attention-based fusion, for automated skin lesion classification. The complete system consists of:

Dermoscopic Image → Preprocessing & Augmentation → ResNet-50 CNN Branch + Transformer Encoder Branch → Attention-Based Fusion → Channel-wise Attention → Classification Head → Malignant/Benign Prediction

The study achieved:

- 98.47% classification accuracy on the ISIC 2020 Melanoma Classification Dataset
- Precision of 0.982, Recall of 0.984, and F1-score of 0.983
- An AUC of 0.994, outperforming VGG16, ResNet50, ViT-Base, and EffNet-ViT baselines
- Improved interpretability through attention-based local–global feature fusion

Although broader multi-dataset validation and extension to multi-class disease categories would further strengthen the work, the research provides a valuable and practically relevant foundation for future hybrid CNN–Transformer systems in medical image classification.

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